

Qatar-1b

R-2655

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250910 · created 2026-08-29T23:47:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460928.6986 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.029
Transit depth (Rp/Rs)^2	2.45 +/- 0.92 [%]
Orbital Inclination (inc)	88.91 +/- 2.02
Airmass coefficient 1 (a1)	1.368 +/- 0.027
Airmass coefficient 2 (a2)	-0.25 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.96 %
Best Comparison Star	#2 - [477, 317]
Optimal Aperture	2.91
Optimal Annulus	12.42
Transit Duration (day)	0.0785 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.029 vs 0.1463 (deviation 0.33σ)
Duration fit vs published	0.0785 d vs 0.0692 d (deviation 2.47σ)
Mid-time O-C	-7.7 min
Depth SNR	4.9
Residual scatter	1.96 %

Light curve

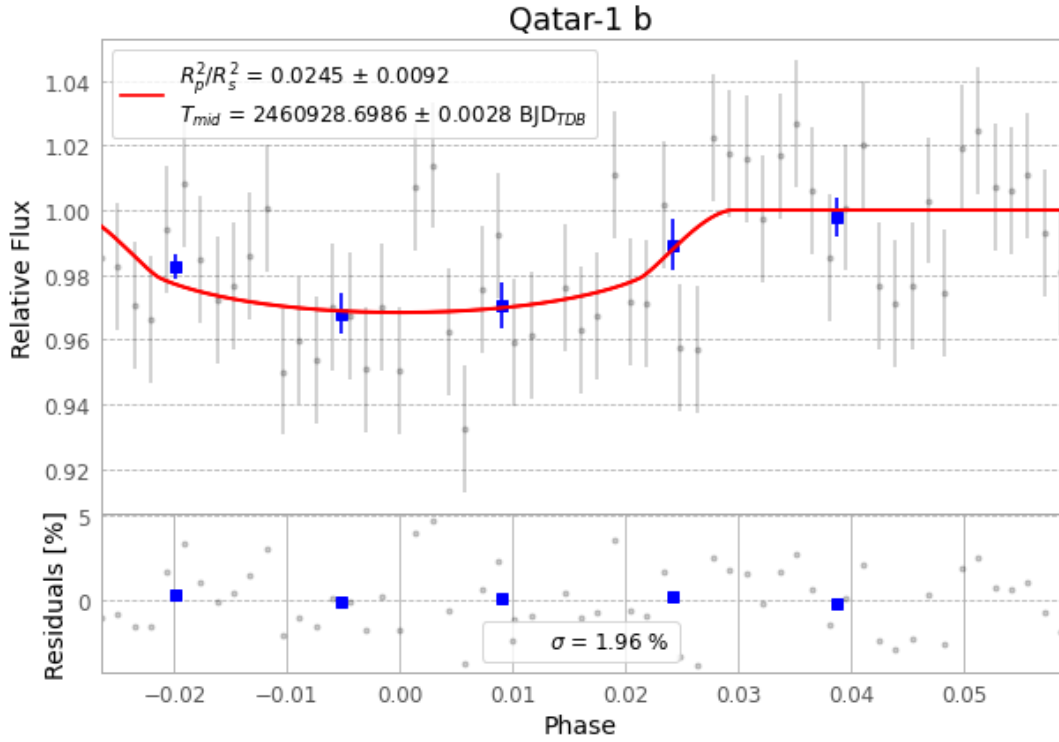


Figure 1 — FinalLightCurve_Qatar-1 b_2025-09-09.png (SHA-256 6dd64be8dc0a6f53...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 791e8ab30951ff0f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-1250910034815.FITS → Qatar-1250910064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250910.log (8d7913089f35...), FinalLightCurve_Qatar-1 b_2025-09-09.png (6dd64be8dc0a...), FinalLightCurve_Qatar-1 b_2025-09-09.csv (e26d236958b2...)

Compute resources

Wall time 153.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:47Z.